Homework #6

Math 742

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Homework: Hypothesis Testing and Likelihood Methods

Problem 1.

Let $X_1, \ldots, X_{25} \overset{iid}{\sim} N(\mu, 1)$. We want to test

$$H_0 : \mu = 0 \quad vs \quad H_1 : \mu > 0$$

Because the alternative says $\mu > 0$, larger values of the sample mean $\bar{x}$ should count as evidence against H_0. So it makes sense to base the test on $\bar{X}$, and then standardize it so that under H_0 it has a known distribution.

(a) Test statistic

Since each $X_i \overset{iid}{\sim} N(\mu, 1)$, the sample mean satisfies $\bar{X} \sim N\left(\mu, \frac{1}{25}\right)$

under the null hypothesis $H_0 : \mu = 0$, this becomes $\bar{X} \sim N\left(0, \frac{1}{25}\right)$

To make the null distribution easy to use, we standardize:

$$Z = \frac{\bar{X} - 0}{1/\sqrt{25}} \longrightarrow \sqrt{25} = 5 \longrightarrow Z = 5\bar{X}$$

Thus, under H_0, $Z \sim N(0,1)$

(b) Rejection region for a level $\alpha = 0.05$ test

Because $H_1 : \mu > 0$, this is a right-tailed test. That means we should reject for large positive values of Z, since those correspond to sample means much larger than 0.

We want a level 0.05 test, so we choose the cutoff c such that

$$P_{H_0}(Z > c) = 0.05$$

Since $Z \sim N(0,1)$ under H_0, the upper 5% cutoff is $c = 1.645$

so the region is Reject H_0 if $Z > 1.645$

Since $Z = 5\bar{X}$, this is equivalent to $5\bar{X} > 1.645$ or $\bar{X} > \frac{1.645}{5} = 0.329$

We can also rewrite the rejection as

$$\boxed{\text{Reject } H_0 \text{ if } \bar{X} > 0.329}$$

(c) p-value when $\bar{X} = 0.32$

Now suppose the observed value (sample mean) is $\bar{X} = 0.32$

Then the observed test statistic is $Z_{obs} = 5(0.32) = 1.60$

Because this is a Right-tailed test, the p-value is the probability, assuming H_0 is true, of seeing a test statistic at least this large: $\text{p-value} = P_{H_0}(Z \geq 1.60)$

Since $Z \sim N(0,1)$, $\text{p-value} = 1 - \Phi(1.60) \longrightarrow \Phi(1.60) \approx 0.9452$ (using the NST)

So, $\text{p-value} \approx 1 - 0.9452 = 0.0548$

Therefore, $\text{p-value} \approx 0.0548$

(d) Interpretation of the p-value

In this setting, the p-value means:

If the true mean were really $\mu = 0$, then the probability of getting a sample mean at least as large as 0.32 from a sample of size 25 is about 0.0548

Equivalently

Assuming H_0 is true, the probability of observing a test statistic at least as extreme as $Z = 1.60$ is about 5.48%

(e) Would the conclusion change if $\alpha = 0.01$?

No, the conclusion would not change.

At significance level $\alpha = 0.01$, we reject only if $\text{p-value} < 0.01$

But here, $0.0548 > 0.01$

So we still **do not reject** H_0.

Equivalently, the 1% upper-tail cutoff is about $Z_{0.09} = 2.326$ and since $1.60 < 2.326$ the observed statistic is not in the rejection region.

Thus,

$$\boxed{\text{At } \alpha = 0.01, \text{ we will do not reject } H_0}$$

Problem 2.

(a) Describe how to construct a level α test using a rejection region of the form $T > c$

Because larger values of $T(x)$ indicate stronger evidence for disease, it makes sense to reject H_0 for large values of T. So we consider the rule

$$\text{Reject } H_0 \text{ if } T > c$$

Now we choose c so that the test has level α. Since under H_0, $T \sim N(0,1)$ we require $P_{H_0}(T > c) = \alpha$. If $Z \sim N(0,1)$, this means $P(Z > c) = \alpha$

Therefore c must be the upper $1-\alpha$ quantile of the standard normal:

$$\boxed{c = \Phi^{-1}(1-\alpha) = z_{1-\alpha}}$$

so the level-α test is

$$\boxed{\text{Reject } H_0 \text{ if } T > z_{1-\alpha}}$$

(b) Express the power at $\mu = 2$ as a function of c

Under H_1, we are told

$$T \mid H_1 \sim N(2,1)$$

The power is the probability of rejecting H_0 when H_1 is true. Since the rejection rule is $T > c$

$$\pi(2) = P(T > c \mid H_1)$$

Because $T \sim N(2,1)$ under H_1, standardize:

$$\pi(2) = P(T > c) = P\left(\frac{T-2}{1} > c-2\right)$$

But

$$\frac{T-2}{1} \sim N(0,1), \quad \text{so} \quad \pi(2) = P(Z > c-2) = 1 - \Phi(c-2)$$

Therefore the power function at the disease model is: $\boxed{\pi(2) = 1 - \Phi(c-2)}$

If we substitute the level-α choice $c = z_{1-\alpha}$, then: $\boxed{\pi(2) = 1 - \Phi(z_{1-\alpha} - 2)}$

(c) Explain the tradeoff between Type I and Type II error in this setting

here:

- Type I error = reject H_0, when there is actually no disease. False Positive.
- Type II error = fail to reject H_0 when disease is actually present. False Negative.

Now think about the cut-off c:

If we make c smaller

Then it is eassier for T to exceed c so we reject more often.

- Type I error increases, because even healthy people will more often cross the threshold.
- Type II error decreases, because diseased people are less likely to be missed.
- Power increases.

If we make c larger

Then it is harder for T to exceed c, so we reject less often.

- Type I error decreases.
- Type II error increases.
- Power decreases.

So the tradeoff is:

$\boxed{\text{Smaller } c \longrightarrow \text{more false positives but fewer false negatives}}$

and

$\boxed{\text{Larger } c \longrightarrow \text{fewer false positives but more false negatives}}$

(d) If the cost of missing a disease is very high, how should c be chosen?

If missing a disease is very costly, then Type II error is much more serious then type I error.

A Type II error here means: fail to detect disease when disease is actually present.

So we want to reduce Type II error as much as possible, which means we want to <u>INCREASE POWER</u>

To do that, we should choose a smaller cutoff c, so that the rejection region $T > c$ becomes larger.

Thus, $\boxed{\text{choose } c \text{ smaller}}$

Problem 3.

Let $S = \sum_{i=1}^{n} X_i$ since the X_i are iid Exponential (λ) with density

$$f(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x > 0,$$

The likelihood is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} X_i} = \lambda^n e^{-\lambda S}$$

We are testing $H_0 : \lambda = 1$ vs $H_1 : \lambda = 2$

(a) The likelihood ratio is $\Lambda(x) = \dfrac{L(\lambda = 1)}{L(\lambda = 2)}$

Compute each likelihood

$$L(1) = 1^n e^{-S} = e^{-S}$$
$$L(2) = 2^n e^{-2S}$$

Therefore,

$$\Lambda(x) = \frac{e^{-S}}{2^n e^{-2S}} = \frac{e^S}{2^n}$$

So,

$$\Lambda(x) = \frac{e^{\sum_{i=1}^{n} X_i}}{2^n}$$

(b) Since $\Lambda(X) = \dfrac{e^S}{2^n}$, where $S = \sum_{i=1}^{n} X_i$ and 2^n is constant, Λ is an increasing function of S. Thus $\Lambda(x)$ is a monotone increasing function of $\sum_{i=1}^{n} X_i$

Recall $\dfrac{d}{dS}\left(\dfrac{e^S}{2^n}\right) = \dfrac{e^S}{2^n} > 0$

This matter because once the likelihood ratio is monotone in a statistic, the region (rejection) can be written directly in terms of that statistic.

(c) For a likelihood-ratio test, we reject H_0 for small values of $\Lambda(x)$. Since $\Lambda(x)$ is increasing in $S = \sum_{i=1}^{n} X_i$, small values of Λ correspond to small values of S.
Therefore the rejection region has the form:

$$\boxed{\text{Reject } H_0 \text{ if } \sum_{i=1}^{n} X_i \leq c}$$

For some constant c. If a level-α test is desired, then c is chosen so that

$$P_{\lambda=1}\left(\sum_{i=1}^{n} X_i \leq c\right) = \alpha$$

This makes sense because the smallest sums are the observations that give the strongest evidence against H_0.

(d) For an Exponential (λ) random variable, $E[X_i] = \frac{1}{\lambda}$
So under $H_0 : \lambda = 1$, $E[X_i] = 1$, while under $H_1 : \lambda = 2$, $E[X_i] = \frac{1}{2}$.
Thus, when $\lambda = 2$, the observations tend to be smaller on average. Therefore, a small value of $\sum_{i=1}^{n} X_i$ is more consistent with $\lambda = 2$ than with $\lambda = 1$. Hence small sums provide evidence for H_1.

Problem 4.

We consider $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ w.

(a) The Neyman-Pearson lemma says that when testing one fully specified null hypothesis against one fully specified alternative hypothesis, and we want to control the type I error at a fixed level α, the best test is the one based on the likelihood ratio. In words, we reject H_0 when the observed data favor H_1 strongly enough relative to H_0. Among all tests with the same size α, this test has the greatest power against H_1. This makes sense because once the false positive rate is fixed, the goal is to maximize the probability of correctly detecting H_1.

(b) A most powerful test is a test that, among all tests with size at most α, has the largest probability of rejecting H_0 when H_1 is true. In other words, after fixing the allowed Type I error rate, it is the test that maximizes power. This makes sense because testing is an optimization problem: control false alarm first, then choose the rule that detects the alternative as often as possible.

(c) The likelihood ratio appears naturally because it directly compares how well the data are explained by H_1 versus H_0. If the likelihood under H_1 is much larger than the likelihood under H_0, then the data are much more compatible with the alternative. The lecture notes emphasize that likelihood ratios are not arbitrary; they arise as the solution to the optimal decision problem of maximizing power while controlling Type I error. Thus the likelihood ratio is the natural quantity to threshold in the optimal test.

(d) In practice, the Neyman-Pearson Lemma is limited because it only applies to simple-vs-simple testing. Most real statistical problems involve composite hypotheses, nuisance parameters, or multiple parameters. In those settings, a test that is best for one alternative value may not be best for another, so a uniformly most powerful test often does not exist.

(e) A real-world situation where the Neyman-Pearson framework is appropiate is diagnostic testing with two fully specified models.

For example: suppose a biomarker has one known distribution in healthy patients and another known distribution in diseased patients. Then the problem is close to a simple null versus a simple alternative, and the likelihood-ratio test gives the most powerful rule at a fixed false positive rate. This makes sense because the two competing models are both completely specified, which is exactly the setting required by the Neyman-Pearson Lemma.

Problem 5.

Let $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, 1)$. We want to test $H_0: \mu = 0$ vs $H_1: \mu > 0$

consider the test that rejects when $\bar{X} > c$

This is a one-sided composite alternative problem. The key ideas (from the notes) are:

- a test is a decision rule based on a statistic and a critical region,
- The power function is $\beta(\theta) = P_\theta(\text{Reject } H_0)$.
- for a one-sided normal-mean test, the power increases as the true mean moves farther into the alternative, and the cutoff is chosen so the test has level α.
- A procedure is consistent when, as the sample size grows, it eventually detects every fixed false null with probability tending to 1. The consistency notes explain this along-run accuracy property driven by the law of large numbers.

(a) Determine c

Since $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, 1)$, $\bar{X} \sim N(\mu, \frac{1}{n})$. under $H_0: \mu = 0$, $\bar{X} \sim N(0, \frac{1}{n})$

For a level-α test, we choose c so that $P_0(\bar{X} > c) = \alpha$

Standarizing under H_0, $P_0(\bar{x} > c) = P\left(\frac{\bar{X}}{1/\sqrt{n}} > \sqrt{n}\, c\right) = P(Z > \sqrt{n}\, c)$,

where $Z \sim N(0,1)$. Therefore we need $\sqrt{n}\, c = Z_{1-\alpha}$, So $\boxed{c = \frac{Z_{1-\alpha}}{\sqrt{n}}}$

(b) The power function is $\beta(\mu) = P_\mu(\text{Reject } H_0) = P_\mu(\bar{x} > c)$

since $\bar{X} \sim N(\mu, \frac{1}{n})$,

$$\beta(\mu) = P_\mu\left(\frac{\bar{x}-\mu}{1/\sqrt{n}} > \sqrt{n}(c-\mu)\right) = P(Z > \sqrt{n}(c-\mu)),$$

So $\beta(\mu) = 1 - \Phi(\sqrt{n}(c-\mu))$

Substituting $c = Z_{1-\alpha}/\sqrt{n}$

$$\boxed{\beta(\mu) = 1 - \Phi(Z_{1-\alpha} - \sqrt{n}\mu)}$$

(c) As μ increases, the quantity $Z_{1-\alpha} - \sqrt{n}\mu$ decreases, so $\Phi(Z_{1-\alpha} - \sqrt{n}\mu)$ decreases, and therefore $\beta(\mu) = 1 - \Phi(Z_{1-\alpha} - \sqrt{n}\mu)$ increases.

Thus the power function is increasing in μ. It equals α at $\mu = 0$, is larger than α for $\mu > 0$, and approaches 1 as μ becomes large. So the power curve starts at the significance level and rises as the true mean moves farther into the alternative.

(d) This test is consistent if for every fixed $\mu > 0$, $\beta_n(\mu) \to 1$ as $n \to \infty$

In words, if the null is false and the true mean is any positive number, then with enough data the probability of rejecting H_0 approaches 1. This matches the general large-sample idea that more data should make the correct decision more likely.

(e) from part (b), $\beta_n(\mu) = 1 - \Phi(Z_{1-\alpha} - \sqrt{n}\mu)$ fix any $\mu > 0$. Then as $n \to \infty$, $\sqrt{n}\mu \to \infty$

so, $Z_{1-\alpha} - \sqrt{n}\mu \to -\infty$. Hence $\Phi(Z_{1-\alpha} - \sqrt{n}\mu) \to 0$, and therefore $\beta_n(\mu) \to 1$

so the test is consistent: $\boxed{\text{Yes, this test is consistent}}$